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Repair-Aware Quantum Sampling for a Discrete Process-Design QUBO

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ABSTRACT

This paper evaluates what is required to interpret gate-based quantum optimization samples for a drug-substance manufacturing discrete-design problem. The exported integer program has 19 binary flow variables, while the QUBO used by the quantum workflows has 36 variables after adding auxiliaries. We show that QAOA and VQE samples become process-design candidates only after projection to the original flow variables and exact auxiliary repair. After projection and repair, the primary non-oracle workflow—energy-targeted $p = 5$ QAOA on a reduced quadratic surrogate—sampled the reference optimum 664 times in 262144 ideal Aer reads. This reference flow is the best retained Gurobi-reported solution, the exact repaired-QUBO minimizer over all 2^{19} projected flows, and the optimum in a local Gurobi 13.0.2 rerun with zero MIPGap. An oracle-aided top-10 diagnostic, using the enumerated repaired ranking, sampled it 1007 times. A historical selected-seed VQE follow-up sampled the same reference optimum 20 times in 524288 reads, while an IBM hardware pilot found one top-10 stored Gurobi-pool flow and no reference-optimum read. The evidence supports repair-aware workflow feasibility, not speedup, hardware superiority, or Gurobi replacement.

Keywords: quantum computing, process design, decomposition, optimization

INTRODUCTION

Discrete choices are often the first barrier in process design. Before a flowsheet can be simulated or optimized continuously, the designer must choose among routes, units, operating modes, and other mutually exclusive configurations [1, 2]. These choices make many process-design models combinatorial. Quantum optimization is attractive in this setting because QAOA and VQE are designed to sample low-energy binary states [3–5]. For chemical engineering and process systems engineering, recent reviews and case studies frame quantum computing as a workflow question rather than a drop-in solver replacement [6–8].

The practical question is not only whether a process-design model can be written as a QUBO [9–11], but whether the resulting samples remain interpretable after reformulation, auxiliary variables, parameter selection, backend configuration, and post-processing are made explicit. The contribution is a repair-aware audit showing how QUBO samples must be mapped, repaired, and scored before comparison with process-design solutions.

We study a drug-substance manufacturing (DS-MFG) discrete-design instance. The original Gurobi model contains 19 binary flow variables and a retained solution pool. The QUBO artifact used by the quantum workflows contains 36 variables because it also contains 17 auxiliary variables. A sampled 36-bit QUBO state is therefore not, by itself, a process-design decision. It must first be projected onto the original 19 flow variables and then scored with the best auxiliary completion. This small but fully enumerable DS-MFG instance is used as a controlled testbed: it is large enough to expose reformulation and sampling issues, but small enough to audit every repaired flow assignment.

The conclusion is conservative. After reduction, exact auxiliary repair, and offline QAOA angle search, local Aer simulations place observable mass on the reference optimum, and a fixed-parameter IBM hardware pilot executes the handoff and scoring path. These results demonstrate feasibility and engineering burden, not a runtime advantage over Gurobi or a hardware-performance advantage over classical simulation.

METHODS

Discrete Design Instance

The original DS-MFG instance is a binary integer-programming model for a capital-expenditure selection problem. The 19 binary variables encode candidate material-flow arcs or unit-operation selections in a manufacturing superstructure, and the objective is a normalized capital-cost score. The reference value is 11.7095; it is the best retained Gurobi-reported value and is certified by the local Gurobi 13.0.2 rerun described below. Schematically, the discrete subproblem has the form

$$\begin{aligned}
\min_{f} \quad & \sum_{i\in F} c_i f_i \\
\text{s.t.} \quad & \sum_{i\in \mathsf{In}(n)} f_i = \sum_{j\in \mathsf{Out}(n)} f_j \qquad \forall n \in N, \\
& f_{\mathsf{source}} = 1, \quad f_{\mathsf{sink}} = 1, \\
& g(f) \le 0, \\
& f_i \in \{0,1\} \qquad\qquad\qquad \forall i \in F.
\end{aligned} \tag{1}$$

Equation (1) records the relevant constraint classes, not a literal listing. The exact Gurobi pool, QUBO coefficients, and repair artifacts are retained in the supplied QUBO archive and the case-study artifact bundle. In the Gurobi export the model is named `capex`; it has 19 binary flow variables, 20 constraints, 58 nonzeros, and a minimization objective. The original retained Gurobi export reports `PoolSearchMode=2`, `PoolSolutions=36`, solve time 0.073 s, and 36 solutions found [12]; it did not retain solver version, final status, MIPGap, PoolGap, or pool intent. A local Gurobi 13.0.2 rerun of the same IP with `PoolSearchMode=2` and `PoolSolutions=100` returned `OPTIMAL`, `MIPGap=0.0`, objective 11.7095, flow `1001110100111100011` in $(f_{00}, \dots, f_{18})$ order, 36 solutions, and exact retained-pool agreement. This classical rerun did not involve quantum hardware; it validates the exported IP and supplies the Gurobi-certified reference. Exact enumeration below verifies the same flow is the repaired-QUBO minimizer over all 2^{19} projected flows. We report whether a sampled flow matches a stored Gurobi-pool solution; this is a pool-membership metric, not a certificate that all other sampled flows are infeasible.

QUBO Representation, Repair, and Surrogate

The supplied QUBO archive provides a scalar file, a linear coefficient vector, and a quadratic coefficient matrix. For a full bitstring $x \in \{0,1\}^{36}$, the encoded objective is

$$E_{\mathsf{QUBO}}(x) = a\left(\ell^\top x + x^\top Q x + b\right), \tag{2}$$

with $a = 1.0$, $b \approx 601.476$, $\ell \in \mathbb{R}^{36}$, and $Q \in \mathbb{R}^{36\times 36}$. The stored matrix is evaluated as exported: off-diagonal interactions are present in the upper-triangular part of Q and are counted once in $x^\top Q x$.

The QUBO variables separate into original flow variables and auxiliaries,

$$x = (y, z), \qquad y \in \{0,1\}^{19}, \quad z \in \{0,1\}^{17}.$$

Every sampled state is scored after exact auxiliary repair:

$$F(y) = \min_{z\in\{0,1\}^{17}} E_{\mathsf{QUBO}}(y, z). \tag{3}$$

Components are connected components of the auxiliary-auxiliary interaction graph after conditioning on y; each component is enumerated exactly. For this instance, fixing y decomposes the repair into 13 components, each containing one or two auxiliary variables.

This repair gives a valid bridge back to the IP objective. The exact repaired minimizer over all 2^{19} flow assignments is the retained Gurobi-reported reference flow 1001110100111100011, ordered as $(f_{00}, \dots, f_{18})$, with objective 11.7095. Across all 36 stored Gurobi-pool flows, repaired QUBO scores match the IP objective within 3.1×10^{-13}. Fourteen of the top 50 repaired flows are not in the stored Gurobi pool, so these non-pool flows are not certified feasible or infeasible by the pool alone.

The exact repaired objective is not generally quadratic in y. To use QAOA and VQE interfaces that expect a QUBO, we fit a 19-variable quadratic surrogate,

$$\text{quadratic surrogate } \hat{F}(y) = \hat{b} + \hat{\ell}^\top y + y^\top \hat{Q} y, \tag{4}$$

by least squares over all 2^{19} repaired flow assignments. Although the surrogate has high global fit quality ($R^2 \approx 0.999$), it is not reliable as a minimizer: the surrogate optimum repairs to 14.5505 rather than the true 11.7095 optimum. The true optimum is only rank 33 under the surrogate, and among the exact top-50 repaired flows the median and maximum absolute surrogate errors are 18.7 and 29.4, respectively. In the lowest 1% of repaired flows, the exact-rank versus surrogate-rank Spearman correlation is 0.901. The surrogate $\hat{F}$ is used only to generate variational samples; all reported sample-quality metrics are evaluated using the exact repaired objective F, not $\hat{F}$.

Quantum, Classical, and Hardware Workflows

The local QAOA and VQE workflow is reproduced through `QiskitOpt.jl`, an unofficial Julia wrapper around Qiskit Optimization and Aer, with `Qiskit Aer` [13–16]. The current committed project and manifest pin `QiskitOpt.jl` v0.7.0 (git tree `a601dccdd43fd0a8b18e63f3b771273817af49eb`), `QUBODrivers.jl` v0.6.1, `QUBOTools.jl` v0.13.0, `PythonCall.jl` v0.9, and Julia 1.10 compatibility. The Python environment pins Qiskit 2.3.1, Qiskit Aer 0.17.2, Qiskit Optimization 0.7.0, and Qiskit IBM Runtime 0.46.1. The repository smoke test validates package imports and cached scoring summaries under these pins; it does not rerun the expensive stochastic campaigns. The most expensive local simulations use Aer's matrix-product-state method, a tensor-network statevector strategy whose cost is controlled by entanglement and bond dimension [17, 18]. The reduced QAOA and VQE revisit scripts set truncation threshold 10^{-10}, maximum MPS bond dimension 64, `mps_heuristic` measurement sampling, single precision, and a thread cap of `min(Sys.CPU_THREADS, 8)`. Whether truncation occurred was not persisted in the cached Aer outputs. The primary simulator-accuracy check for the central QAOA row is an independent JuliQAOA statevector calculation: it predicted reference-optimum probability 2.63×10^{-3}, while the cached Aer MPS sampled rate was 2.53×10^{-3} with Wilson 95% interval $[2.35, 2.73] \times 10^{-3}$.

Several workflows were evaluated. Direct full-QUBO QAOA and VQE used the 36-variable formulation. Reduced-surrogate QAOA and VQE used the 19-flow surrogate and were scored by exact repair. The highest-concentration QAOA workflow in this audit added an offline angle-search step: `JuliQAOA.jl` was used to optimize QAOA angles over the reduced surrogate [19], and those fixed angles were then transferred back to `QiskitOpt.QAOA` with `MaximumIterations() = 0`. The energy-targeted angle search used basin hopping through `find_angles_bh` with seed 91001 and five basin-hopping iterations, minimizing the expected surrogate energy

$$\mathcal{J}_{\text{energy}}(\theta) = \mathbb{E}_{y \sim p_\theta}\left[\text{quadratic surrogate } \hat{F}(y)\right]. \tag{5}$$

The top-10-targeted diagnostic maximized

$$\mathcal{J}_{10}(\theta) = \sum_{y \in \mathcal{T}_{10}} p_\theta(y), \tag{6}$$

where $\mathcal{T}_{10}$ is the enumerated exact top-10 repaired-flow set. This top-10 objective is oracle-aided because the repaired ranking is known for this small enumerated instance; it is not claimed as a scalable parameter-setting strategy. Reduced-surrogate VQE used `EfficientSU2`, 152 initial parameters on the 19-flow surrogate, 128 optimizer reads, 25 COBYLA iterations, and final-read sweeps up to 524288 reads. The main VQE row is a historical cached selected-seed follow-up: seed 74018 produced 20 reference-optimum hits in 524288 reads. A later metadata rerun produced 4 reference-optimum hits in 1572864 reads and documents provenance without replacing the historical distribution.

Classical baselines sampled the repaired 19-flow objective by uniform random sampling and by steepest-descent single-bit-flip hill climbing with random restarts. The hardware pilot used the top-10-targeted $p = 5$ fixed QAOA circuit as a handoff test on `ibm_fez`. The hardware manifest records the backend query timestamp (2026-06-17 20:19 UTC), 156-qubit backend, basis gates, 19-qubit logical circuit, pre-transpile depth 86, 255 `rzz` operations, 4096 shots per submitted job, three repeats, transpile seeds 92001–92003, and nine completed job IDs. It does not record a calibration snapshot, queue time, readout mitigation, or error mitigation; no such mitigation is claimed.

Evaluation Metrics

For a sampled QUBO state $x = (y, z)$, we record the raw QUBO energy $E_{\text{QUBO}}(y, z)$, the repaired objective $F(y)$, and the auxiliary gap $E_{\text{QUBO}}(y, z) - F(y)$. A reference-optimum hit means that the projected y equals the reference-optimal 19-bit flow after exact auxiliary repair; it does not require the sampled 36-bit string to contain the optimal auxiliary completion. The main events are reference-optimum hits, top-10 and top-50 repaired-rank hits, and stored Gurobi-pool hits. Empirical hit rates are accompanied in the artifact tables by Wilson intervals and empirical 99% time-to-target diagnostics [20]. These read-level or evaluation-level diagnostics exclude exact enumeration, surrogate fitting, QAOA angle search, transpilation, queueing, hardware calibration, and repair-cache construction.

RESULTS

Repair Changes the Meaning of a Sample

Directly sampling the 36-variable QUBO was not enough to obtain an interpretable process-design result. In an early full-QUBO QAOA final-sampling run, $p = 2$ QAOA produced no reference-optimum reads in 512 final samples, although exact repair recovered a best stored-pool flow with repaired objective 14.5505. A higher-read direct full-QUBO run later sampled the repaired reference optimum 4 times in 32768 reads, but these were repaired-flow hits rather than encoded 36-bit optimum completions. Figure 1 summarizes the resulting scoring path: the original flow variables define the process-design candidate, and auxiliary repair defines the score used for comparison. These direct 36-bit runs are not used as performance claims; they show that full-QUBO samples are difficult to interpret without projection and repair.

Reduction and Angle Transfer Concentrate QAOA Samples

The primary non-oracle QAOA result is the energy-targeted $p = 5$ transferred workflow. Here energy-targeted means minimizing the expected reduced-surrogate energy in Eq. (5), not using the exact repaired top-k ranking. This workflow sampled the reference optimum 664 times in 262144 ideal Aer reads, for an empirical reference-hit rate of 2.53×10^{-3} with Wilson 95% interval $[2.35, 2.73] \times 10^{-3}$. It also produced 39661 top-50 reads, 8964 top-10 reads, and 32502 stored Gurobi-pool reads.

The top-10-targeted $p = 5$ angle objective improved the local diagnostic result to 1007 reference-optimum reads in the same 262144-read budget, together with 62597 top-50 reads and 14326 top-10 reads. That result is useful for stress testing the circuit and hardware handoff, but it is oracle-aided because it uses the exact repaired ranking obtained by enumerating all 2^{19} flow assignments.

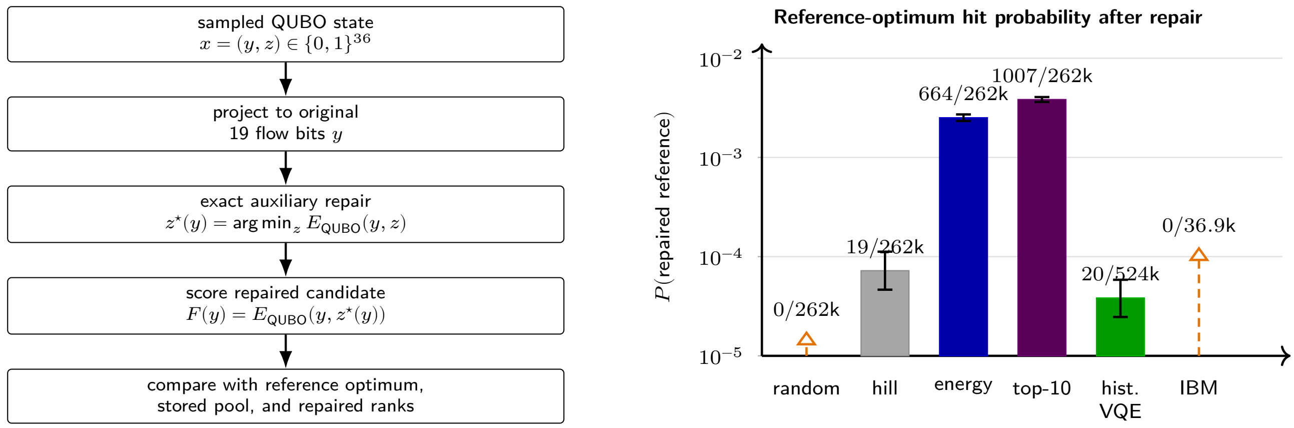

Figure 1. Repair-aware scoring pipeline and representative reference-optimum hit rates after projection to the 19 original flow variables and exact auxiliary repair. Solid bars are observed probabilities with Wilson 95% intervals; open triangles are Wilson upper bounds for zero-hit rows, not observed hits. Short labels denote uniform random sampling, hill-climb restarts, energy-targeted transferred QAOA, oracle-aided top-10-targeted QAOA, historical selected-seed VQE follow-up, and the `ibm_fez` hardware pilot.

VQE and Classical Baselines Bound the Claim

VQE also benefited from reduction, but it produced a less concentrated distribution. Full-QUBO VQE did not sample the reference optimum in the five-seed 8192-read sweep. Reduced-surrogate VQE found a near-optimal stored-pool flow with objective 11.8105 in a five-seed sweep and sampled the reference optimum three times in a larger 20-seed sweep. Seed 74018 was selected for the final high-read follow-up because the preceding 262144-read selected-seed run had the largest top-50, top-10, and reference-hit counts among the selected seeds. The historical final 524288-read follow-up for seed 74018 produced 20 reference-optimum reads, 152 top-10 reads, and 391 top-50 reads.

The classical baselines prevent overclaiming. Uniform random repaired-flow sampling did not find the reference optimum in either 262144 or 524288 samples, but this is consistent with chance: with one optimum among 2^{19} flows, the expected count in 262144 uniform samples is 0.5. Hill climbing with random restarts was much stronger: with 262144 repaired-objective evaluations, it found the reference optimum 19 times and reached 883 top-50 evaluations. The comparison is distributional: at matched nominal counts, QAOA places more mass on repaired low-rank flows than these simple baselines, but the counts do not normalize preprocessing, angle search, simulation, or solver overhead.

Hardware Execution Is a Feasibility Handoff

The fixed-parameter top-10-targeted QAOA circuit was submitted to `ibm_fez` using three repeats and transpile seeds 92001–92003, with 4096 shots per repeat/seed pair. All 9 submitted jobs completed, for 36864 total hardware reads. The hardware pilot observed 6 top-50 reads, 1 top-10 read, 4 stored Gurobi-pool reads, and 0 reference-optimum reads. The best hardware sample was the rank-2 repaired flow with objective 11.8105. With zero reference hits, the Wilson 95% upper bound on the hardware reference-hit rate is about 1.04×10^{-4}.

Because calibration, mitigation, and full submitted-job transpilation metadata were not retained, the hardware result is used only to validate the submission-and-scoring path, not to diagnose a specific noise mechanism. It shows a lower observed hit rate than the ideal simulator in this run, but the retained metadata are insufficient to attribute the difference to a specific noise source. The fake-backend circuits had depths 1442–1557 and 1136–1156 `cz` gates, but the fake backend is not a calibrated predictor for the executed hardware jobs. These observations make the hardware pilot useful as a handoff exercise, not as evidence of hardware superiority.

DISCUSSION AND LIMITATIONS

This study answers a narrower but more useful question than whether quantum optimization "solves" the DS-MFG instance. The instance already has a retained Gurobi reference solution, a matching local Gurobi rerun, and an exact repaired-QUBO minimizer over the projected flow space. The relevant contribution is identifying what has to happen for a QAOA or VQE distribution to produce meaningful process-design samples. The answer is not a drop-in solver replacement: it requires formulation audit, separation of original and auxiliary variables, exact repair, reduced-surrogate construction, tuned parameter search, backend-specific execution choices, and postprocessing against the original Gurobi flow space. The positive result is concentration of a repaired-flow sampling distribution after substantial

Table 1: Primary outcomes in the manuscript body. All stochastic rows are scored after exact auxiliary repair. The top-10-targeted QAOA row is an oracle-aided diagnostic, not the primary scalable parameter-setting claim.

Workflow	Budget	Reference-hit rate	Top-10 / top-50 reads	Interpretation
Gurobi solution pool	Deterministic IP solve; local rerun	Defines reference	–	Objective 11.7095 and flow `1001110100111100011`; local Gurobi 13.0.2 rerun returned `OPTIMAL` with `MIPGap=0.0`.
Uniform random repaired-flow sampling	262144 samples	0	5 / 24	Lower concentration baseline; best sampled rank 2 with objective 11.8105.
Hill-climb random restarts	262144 evaluations	$7.25{\times}10^{-5}$	177 / 883	Strong simple classical baseline once exact repair is available.
Energy-targeted transferred QAOA	262144 Aer reads	$2.53{\times}10^{-3}$	8964 / 39661	Primary non-oracle QAOA result; 664 reference reads.
Top-10-targeted transferred QAOA	262144 Aer reads	$3.84{\times}10^{-3}$	14326 / 62597	Oracle-aided diagnostic using the enumerated repaired ranking; 1007 reference reads.
Historical reduced-surrogate VQE, seed 74018	524288 Aer reads	$3.81{\times}10^{-5}$	152 / 391	Reached the optimum, but with lower concentration than transferred QAOA; not replaced by the current metadata rerun.
IBM `ibm_fez` QAOA pilot	36864 hardware reads	0	1 / 6	Executed the fixed-circuit handoff; best sampled flow was rank 2 with objective 11.8105.

preprocessing, not an end-to-end optimization advantage.

Because the repaired 19-flow landscape is enumerable in this case, the study should be read as a validation and workflow-audit testbed rather than a scalable optimization demonstration. The top-10-targeted QAOA diagnostic uses knowledge of the repaired low-energy set, reported time-to-target values exclude preprocessing and tuning overheads, and stored Gurobi-pool membership is not a complete feasibility certificate unless the pool is exhaustive. The VQE workflow should also persist optimized ansatz parameters so that any future hardware experiment can separate classical optimization from fixed-parameter hardware sampling. The artifact package supports inspection and rescoring of retained samples and summaries; fully rerunning every stochastic campaign would require optimizer-state, simulator-log, and submitted-job hardware metadata that are incomplete for some historical runs.

CONCLUSION

For this DS-MFG case study, QAOA and VQE become interpretable only after the QUBO auxiliary variables are separated from the original process-design variables and repaired exactly. With that bridge in place, energy-targeted transferred-angle QAOA sampled the reference optimum 664 times in 262144 local Aer reads, and an oracle-aided top-10 diagnostic sampled it 1007 times. The historical selected-seed reduced-surrogate VQE follow-up sampled the same reference optimum 20 times in 524288 reads; a current metadata rerun is retained for optimized-parameter provenance rather than replacing that cached result. A small IBM hardware pilot executed the QAOA handoff and found a near-optimal stored-pool flow, but no reference-optimum read.

The main lesson is that quantum optimization samples become process-design candidates only after projection, auxiliary repair, and scoring in the original decision space. The evidence supports feasibility and interpretation, not replacement of Gurobi or a claim of quantum speedup.

ARTIFACT AVAILABILITY

Detailed audit tables, campaign summaries, repair components, exact reduced-objective data, QAOA and VQE distributions, classical baselines, time-to-target reports, hardware manifests, and current VQE metadata are retained in the artifact bundle and in `MANUSCRIPT_FINDINGS.tex`; selected ansatz vectors are in `ds_mfg_vqe_reduced_flow_objective_selected_metadata/`. The repository is https://github.com/SECQUOIA/pse-quantum-discrete-optimization. A final archival DOI will be added before camera-ready submission; the review artifact is identified by the repository snapshot that contains this manuscript and by the checksums in the supplement. Checksums and validation anchors are listed in `case-studies/ds-mfg-qubo-qiskitopt/supplement_standalone.tex`. The bundle supports rescoring retained samples, summaries, repair tables, and validation outputs; byte-for-byte replay is not claimed because some optimizer states, MPS logs, and submitted-job metadata are incomplete.

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